

# MATHEMATICS INFORMATIQUE

Upper Secondary / Terminale Computing Mathematics

## COURSE FOCUS

- Discrete mathematics and logic
- Probability, statistics and numerical methods
- Graphs, matrices and optimization
- Mathematical foundations for algorithms and computing

**80-PAGE COMPLETE COURSEBOOK**

GETI learning resource • structured study • practice • revision • reference

# How to Use This Book

## WHAT YOU WILL LEARN

- Core concepts and methods across mathematics informatique.
- Clear explanations, model applications and guided practice.
- Thirty complete lessons followed by mixed revision tests.
- Answer guidance, quick-reference material and final revision checklist.

## RECOMMENDED STUDY METHOD

1. Read the lesson objective before studying the main idea.
2. Study the key terms and model example, then explain them in your own words.
3. Complete the practice page without checking the answer guidance.
4. Correct mistakes by writing the rule, principle or method behind the correction.
5. Review difficult lessons after 24 hours and again after one week.

## BOOK ORGANIZATION

- Lessons 1-15: foundations and core methods.
- Lessons 16-30: advanced applications and integrated study.
- Pages 65-72: eight mixed revision tests.
- Pages 73-78: answer guidance and correction strategies.
- Pages 79-80: quick reference and final checklist.

# Table of Contents

## Lessons 1-15

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| 4      | Proof Techniques          | 11-12 |
| 5      | Combinatorics             | 13-14 |
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### LEARNING MAP

Study the course progressively. Each lesson has a concept page followed by a practice page. The final section contains review tests, answer guidance and quick reference pages.

# Table of Contents

## Lessons 16-30

| Lesson | Topic                                       | Pages |
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| 18     | Numerical Approximation                     | 39-40 |
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### LEARNING MAP

Study the course progressively. Each lesson has a concept page followed by a practice page. The final section contains review tests, answer guidance and quick reference pages.

# Logic and Propositions

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of logic and propositions.
- Apply logic and propositions to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Logic and Propositions is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of logic and propositions.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain logic and propositions in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Logic and Propositions - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define logic and propositions in your own words.
2. List three key ideas connected to logic and propositions.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of logic and propositions. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Sets

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of sets.
- Apply sets to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Sets is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of sets.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain sets in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Sets - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define sets in your own words.
2. List three key ideas connected to sets.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of sets. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Relations and Functions

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of relations and functions.
- Apply relations and functions to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Relations and Functions is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of relations and functions.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain relations and functions in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Relations and Functions - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define relations and functions in your own words.
2. List three key ideas connected to relations and functions.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of relations and functions. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Proof Techniques

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of proof techniques.
- Apply proof techniques to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Proof Techniques is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of proof techniques.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain proof techniques in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Proof Techniques - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define proof techniques in your own words.
2. List three key ideas connected to proof techniques.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of proof techniques. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Combinatorics

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of combinatorics.
- Apply combinatorics to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Combinatorics is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of combinatorics.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain combinatorics in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Combinatorics - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define combinatorics in your own words.
2. List three key ideas connected to combinatorics.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of combinatorics. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Boolean Algebra

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of boolean algebra.
- Apply boolean algebra to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Boolean Algebra is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of boolean algebra.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain boolean algebra in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Boolean Algebra - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define boolean algebra in your own words.
2. List three key ideas connected to boolean algebra.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of boolean algebra. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Binary and Number Systems

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of binary and number systems.
- Apply binary and number systems to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Binary and Number Systems is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of binary and number systems.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain binary and number systems in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Binary and Number Systems - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define binary and number systems in your own words.
2. List three key ideas connected to binary and number systems.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of binary and number systems. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Modular Arithmetic

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of modular arithmetic.
- Apply modular arithmetic to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Modular Arithmetic is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of modular arithmetic.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain modular arithmetic in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Modular Arithmetic - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define modular arithmetic in your own words.
2. List three key ideas connected to modular arithmetic.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of modular arithmetic. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Sequences and Recurrence

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of sequences and recurrence.
- Apply sequences and recurrence to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Sequences and Recurrence is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of sequences and recurrence.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain sequences and recurrence in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Sequences and Recurrence - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define sequences and recurrence in your own words.
2. List three key ideas connected to sequences and recurrence.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of sequences and recurrence. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Matrices

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of matrices.
- Apply matrices to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Matrices is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of matrices.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain matrices in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Matrices - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define matrices in your own words.
2. List three key ideas connected to matrices.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of matrices. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Vectors

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of vectors.
- Apply vectors to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Vectors is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of vectors.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain vectors in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Vectors - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define vectors in your own words.
2. List three key ideas connected to vectors.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of vectors. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Graph Theory

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of graph theory.
- Apply graph theory to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Graph Theory is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of graph theory.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain graph theory in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Graph Theory - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define graph theory in your own words.
2. List three key ideas connected to graph theory.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of graph theory. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Trees

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of trees.
- Apply trees to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Trees is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of trees.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain trees in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Trees - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define trees in your own words.
2. List three key ideas connected to trees.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of trees. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Discrete Probability

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of discrete probability.
- Apply discrete probability to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Discrete Probability is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of discrete probability.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain discrete probability in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Discrete Probability - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define discrete probability in your own words.
2. List three key ideas connected to discrete probability.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of discrete probability. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Random Variables

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of random variables.
- Apply random variables to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Random Variables is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of random variables.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain random variables in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Random Variables - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define random variables in your own words.
2. List three key ideas connected to random variables.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of random variables. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Statistics for Computing

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of statistics for computing.
- Apply statistics for computing to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Statistics for Computing is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of statistics for computing.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain statistics for computing in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Statistics for Computing - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define statistics for computing in your own words.
2. List three key ideas connected to statistics for computing.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of statistics for computing. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Linear Regression

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of linear regression.
- Apply linear regression to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Linear Regression is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of linear regression.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain linear regression in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Linear Regression - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define linear regression in your own words.
2. List three key ideas connected to linear regression.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of linear regression. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Numerical Approximation

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of numerical approximation.
- Apply numerical approximation to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Numerical Approximation is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of numerical approximation.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain numerical approximation in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Numerical Approximation - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define numerical approximation in your own words.
2. List three key ideas connected to numerical approximation.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of numerical approximation. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Roots of Equations

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of roots of equations.
- Apply roots of equations to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Roots of Equations is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of roots of equations.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain roots of equations in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Roots of Equations - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define roots of equations in your own words.
2. List three key ideas connected to roots of equations.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of roots of equations. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Interpolation

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of interpolation.
- Apply interpolation to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Interpolation is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of interpolation.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain interpolation in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Interpolation - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define interpolation in your own words.
2. List three key ideas connected to interpolation.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of interpolation. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Numerical Differentiation and Integration

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of numerical differentiation and integration.
- Apply numerical differentiation and integration to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Numerical Differentiation and Integration is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of numerical differentiation and integration.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain numerical differentiation and integration in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Numerical Differentiation and Integration - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define numerical differentiation and integration in your own words.
2. List three key ideas connected to numerical differentiation and integration.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of numerical differentiation and integration. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Algorithm Complexity Mathematics

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of algorithm complexity mathematics.
- Apply algorithm complexity mathematics to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Algorithm Complexity Mathematics is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of algorithm complexity mathematics.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain algorithm complexity mathematics in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Algorithm Complexity Mathematics - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define algorithm complexity mathematics in your own words.
2. List three key ideas connected to algorithm complexity mathematics.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of algorithm complexity mathematics. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Finite State Models

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of finite state models.
- Apply finite state models to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Finite State Models is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of finite state models.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain finite state models in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Finite State Models - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define finite state models in your own words.
2. List three key ideas connected to finite state models.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of finite state models. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Linear Optimization

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of linear optimization.
- Apply linear optimization to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Linear Optimization is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of linear optimization.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain linear optimization in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Linear Optimization - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define linear optimization in your own words.
2. List three key ideas connected to linear optimization.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of linear optimization. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Cryptographic Mathematics: Foundations

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of cryptographic mathematics: foundations.
- Apply cryptographic mathematics: foundations to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Cryptographic Mathematics: Foundations is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of cryptographic mathematics: foundations.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain cryptographic mathematics: foundations in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Cryptographic Mathematics: Foundations - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define cryptographic mathematics: foundations in your own words.
2. List three key ideas connected to cryptographic mathematics: foundations.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of cryptographic mathematics: foundations. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Coding and Error Detection

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of coding and error detection.
- Apply coding and error detection to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Coding and Error Detection is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of coding and error detection.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain coding and error detection in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Coding and Error Detection - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define coding and error detection in your own words.
2. List three key ideas connected to coding and error detection.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of coding and error detection. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Data Representation Mathematics

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of data representation mathematics.
- Apply data representation mathematics to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Data Representation Mathematics is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of data representation mathematics.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain data representation mathematics in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Data Representation Mathematics - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define data representation mathematics in your own words.
2. List three key ideas connected to data representation mathematics.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of data representation mathematics. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Computational Geometry

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of computational geometry.
- Apply computational geometry to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Computational Geometry is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of computational geometry.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain computational geometry in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Computational Geometry - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define computational geometry in your own words.
2. List three key ideas connected to computational geometry.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of computational geometry. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Simulation and Mathematical Modelling

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of simulation and mathematical modelling.
- Apply simulation and mathematical modelling to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Simulation and Mathematical Modelling is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of simulation and mathematical modelling.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain simulation and mathematical modelling in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Simulation and Mathematical Modelling - Practice

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define simulation and mathematical modelling in your own words.
2. List three key ideas connected to simulation and mathematical modelling.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of simulation and mathematical modelling. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Integrated Mathematics for Computing Review

Concepts, methods, model examples and structured explanation

## LEARNING OBJECTIVES

- Explain the central ideas and vocabulary of integrated mathematics for computing review.
- Apply integrated mathematics for computing review to a clear academic, technical or real-life example.
- Use a structured method to analyse, solve or evaluate a related problem.

## CORE EXPLANATION

Integrated Mathematics for Computing Review is an important part of MATHEMATICS INFORMATIQUE. A strong learner should first identify the main concept, define the relevant terms precisely, then connect the idea to evidence, a model, a calculation, a system, or a practical situation. The aim is not only to memorize a definition but to understand how the concept works, what assumptions it uses, and how a result can be checked.

### KEY POINTS

- Identify the purpose and structure of integrated mathematics for computing review.
- Separate definitions, procedures, evidence and conclusions.
- Use examples that directly match the concept rather than unrelated facts.
- Check accuracy, units, assumptions, sources or logical consistency where relevant.

## MODEL APPLICATION

Model task: explain integrated mathematics for computing review in a short response. Start with a precise definition, identify the key mechanism or method, apply it to one concrete example, and finish by stating what the example demonstrates.

## STUDY ROUTINE

Step 1: identify the question. Step 2: choose the correct concept or method. Step 3: show reasoning clearly. Step 4: check the result. Step 5: explain the meaning of the answer.

# Integrated Mathematics for Computing Review - Pra

## COMMON ERRORS AND BETTER METHODS

**Common error: Using a vague definition.**

Better: State the exact meaning and the conditions in which it applies.

**Common error: Giving an example without explanation.**

Better: Explain why the example demonstrates the concept.

**Common error: Skipping the checking stage.**

Better: Verify logic, calculations, evidence or assumptions before concluding.

## GUIDED PRACTICE

1. Define integrated mathematics for computing review in your own words.
2. List three key ideas connected to integrated mathematics for computing review.
3. Give one accurate example and explain why it is relevant.
4. Identify one common error or misconception.
5. Explain one method for checking an answer or conclusion.
6. Compare two related ideas or approaches.
7. Write a short application to school, work, society, science or technology.
8. Write one exam-style question and outline a strong answer.

## MASTERY CHALLENGE

Write a 120-180 word explanation of integrated mathematics for computing review. Include a definition, one example, one limitation or checking step, and a concluding sentence.

# Revision Test 1

## MIXED REVIEW QUESTIONS

1. Explain one central idea from Logic and Propositions.
2. Apply Logic and Propositions to one original example.
3. Explain one central idea from Sets.
4. Apply Sets to one original example.
5. Explain one central idea from Relations and Functions.
6. Apply Relations and Functions to one original example.
7. Explain one central idea from Proof Techniques.
8. Apply Proof Techniques to one original example.
9. Explain one central idea from Combinatorics.
10. Apply Combinatorics to one original example.
11. Explain one central idea from Boolean Algebra.
12. Apply Boolean Algebra to one original example.

## EXAM STRATEGY

Answer the exact command word. Define before evaluating. Show steps for calculations or procedures. Use evidence or examples. Check the final answer for accuracy, relevance and completeness.

# Revision Test 2

## MIXED REVIEW QUESTIONS

1. Explain one central idea from Combinatorics.
2. Apply Combinatorics to one original example.
3. Explain one central idea from Boolean Algebra.
4. Apply Boolean Algebra to one original example.
5. Explain one central idea from Binary and Number Systems.
6. Apply Binary and Number Systems to one original example.
7. Explain one central idea from Modular Arithmetic.
8. Apply Modular Arithmetic to one original example.
9. Explain one central idea from Sequences and Recurrence.
10. Apply Sequences and Recurrence to one original example.
11. Explain one central idea from Matrices.
12. Apply Matrices to one original example.

## EXAM STRATEGY

Answer the exact command word. Define before evaluating. Show steps for calculations or procedures. Use evidence or examples. Check the final answer for accuracy, relevance and completeness.

# Revision Test 3

## MIXED REVIEW QUESTIONS

1. Explain one central idea from Sequences and Recurrence.
2. Apply Sequences and Recurrence to one original example.
3. Explain one central idea from Matrices.
4. Apply Matrices to one original example.
5. Explain one central idea from Vectors.
6. Apply Vectors to one original example.
7. Explain one central idea from Graph Theory.
8. Apply Graph Theory to one original example.
9. Explain one central idea from Trees.
10. Apply Trees to one original example.
11. Explain one central idea from Discrete Probability.
12. Apply Discrete Probability to one original example.

## EXAM STRATEGY

Answer the exact command word. Define before evaluating. Show steps for calculations or procedures. Use evidence or examples. Check the final answer for accuracy, relevance and completeness.

# Revision Test 4

## MIXED REVIEW QUESTIONS

1. Explain one central idea from Trees.
2. Apply Trees to one original example.
3. Explain one central idea from Discrete Probability.
4. Apply Discrete Probability to one original example.
5. Explain one central idea from Random Variables.
6. Apply Random Variables to one original example.
7. Explain one central idea from Statistics for Computing.
8. Apply Statistics for Computing to one original example.
9. Explain one central idea from Linear Regression.
10. Apply Linear Regression to one original example.
11. Explain one central idea from Numerical Approximation.
12. Apply Numerical Approximation to one original example.

## EXAM STRATEGY

Answer the exact command word. Define before evaluating. Show steps for calculations or procedures. Use evidence or examples. Check the final answer for accuracy, relevance and completeness.

# Revision Test 5

## MIXED REVIEW QUESTIONS

1. Explain one central idea from Linear Regression.
2. Apply Linear Regression to one original example.
3. Explain one central idea from Numerical Approximation.
4. Apply Numerical Approximation to one original example.
5. Explain one central idea from Roots of Equations.
6. Apply Roots of Equations to one original example.
7. Explain one central idea from Interpolation.
8. Apply Interpolation to one original example.
9. Explain one central idea from Numerical Differentiation and Integration.
10. Apply Numerical Differentiation and Integration to one original example.
11. Explain one central idea from Algorithm Complexity Mathematics.
12. Apply Algorithm Complexity Mathematics to one original example.

## EXAM STRATEGY

Answer the exact command word. Define before evaluating. Show steps for calculations or procedures. Use evidence or examples. Check the final answer for accuracy, relevance and completeness.

# Revision Test 6

## MIXED REVIEW QUESTIONS

1. Explain one central idea from Numerical Differentiation and Integration.
2. Apply Numerical Differentiation and Integration to one original example.
3. Explain one central idea from Algorithm Complexity Mathematics.
4. Apply Algorithm Complexity Mathematics to one original example.
5. Explain one central idea from Finite State Models.
6. Apply Finite State Models to one original example.
7. Explain one central idea from Linear Optimization.
8. Apply Linear Optimization to one original example.
9. Explain one central idea from Cryptographic Mathematics: Foundations.
10. Apply Cryptographic Mathematics: Foundations to one original example.
11. Explain one central idea from Coding and Error Detection.
12. Apply Coding and Error Detection to one original example.

## EXAM STRATEGY

Answer the exact command word. Define before evaluating. Show steps for calculations or procedures. Use evidence or examples. Check the final answer for accuracy, relevance and completeness.

# Revision Test 7

## MIXED REVIEW QUESTIONS

1. Explain one central idea from Cryptographic Mathematics: Foundations.
2. Apply Cryptographic Mathematics: Foundations to one original example.
3. Explain one central idea from Coding and Error Detection.
4. Apply Coding and Error Detection to one original example.
5. Explain one central idea from Data Representation Mathematics.
6. Apply Data Representation Mathematics to one original example.
7. Explain one central idea from Computational Geometry.
8. Apply Computational Geometry to one original example.
9. Explain one central idea from Simulation and Mathematical Modelling.
10. Apply Simulation and Mathematical Modelling to one original example.
11. Explain one central idea from Integrated Mathematics for Computing Review.
12. Apply Integrated Mathematics for Computing Review to one original example.

## EXAM STRATEGY

Answer the exact command word. Define before evaluating. Show steps for calculations or procedures. Use evidence or examples. Check the final answer for accuracy, relevance and completeness.

# Revision Test 8

## MIXED REVIEW QUESTIONS

1. Explain one central idea from Simulation and Mathematical Modelling.
2. Apply Simulation and Mathematical Modelling to one original example.
3. Explain one central idea from Integrated Mathematics for Computing Review.
4. Apply Integrated Mathematics for Computing Review to one original example.
5. Explain one central idea from Logic and Propositions.
6. Apply Logic and Propositions to one original example.
7. Explain one central idea from Sets.
8. Apply Sets to one original example.
9. Explain one central idea from Relations and Functions.
10. Apply Relations and Functions to one original example.
11. Explain one central idea from Proof Techniques.
12. Apply Proof Techniques to one original example.

## EXAM STRATEGY

Answer the exact command word. Define before evaluating. Show steps for calculations or procedures. Use evidence or examples. Check the final answer for accuracy, relevance and completeness.

# Answer Guidance 1

## HOW TO CHECK YOUR WORK

- Definition questions: use precise subject vocabulary and avoid circular definitions.
- Explanation questions: show cause-and-effect or the sequence of reasoning.
- Comparison questions: use the same criteria for both sides and state similarities and differences.
- Application questions: use details from the scenario, data, system or example.
- Calculation questions: show formulas, substitutions, units and a reasonableness check.
- Evaluation questions: present relevant evidence, limitations and a supported conclusion.
- Technical questions: distinguish components, processes, inputs, outputs and controls.
- Essay questions: organize the answer with a direct introduction, developed paragraphs and a concise conclusion.

# Answer Guidance 2

## HOW TO CHECK YOUR WORK

- Definition questions: use precise subject vocabulary and avoid circular definitions.
- Explanation questions: show cause-and-effect or the sequence of reasoning.
- Comparison questions: use the same criteria for both sides and state similarities and differences.
- Application questions: use details from the scenario, data, system or example.
- Calculation questions: show formulas, substitutions, units and a reasonableness check.
- Evaluation questions: present relevant evidence, limitations and a supported conclusion.
- Technical questions: distinguish components, processes, inputs, outputs and controls.
- Essay questions: organize the answer with a direct introduction, developed paragraphs and a concise conclusion.

# Answer Guidance 3

## HOW TO CHECK YOUR WORK

- Definition questions: use precise subject vocabulary and avoid circular definitions.
- Explanation questions: show cause-and-effect or the sequence of reasoning.
- Comparison questions: use the same criteria for both sides and state similarities and differences.
- Application questions: use details from the scenario, data, system or example.
- Calculation questions: show formulas, substitutions, units and a reasonableness check.
- Evaluation questions: present relevant evidence, limitations and a supported conclusion.
- Technical questions: distinguish components, processes, inputs, outputs and controls.
- Essay questions: organize the answer with a direct introduction, developed paragraphs and a concise conclusion.

# Answer Guidance 4

## HOW TO CHECK YOUR WORK

- Definition questions: use precise subject vocabulary and avoid circular definitions.
- Explanation questions: show cause-and-effect or the sequence of reasoning.
- Comparison questions: use the same criteria for both sides and state similarities and differences.
- Application questions: use details from the scenario, data, system or example.
- Calculation questions: show formulas, substitutions, units and a reasonableness check.
- Evaluation questions: present relevant evidence, limitations and a supported conclusion.
- Technical questions: distinguish components, processes, inputs, outputs and controls.
- Essay questions: organize the answer with a direct introduction, developed paragraphs and a concise conclusion.

# Answer Guidance 5

## HOW TO CHECK YOUR WORK

- Definition questions: use precise subject vocabulary and avoid circular definitions.
- Explanation questions: show cause-and-effect or the sequence of reasoning.
- Comparison questions: use the same criteria for both sides and state similarities and differences.
- Application questions: use details from the scenario, data, system or example.
- Calculation questions: show formulas, substitutions, units and a reasonableness check.
- Evaluation questions: present relevant evidence, limitations and a supported conclusion.
- Technical questions: distinguish components, processes, inputs, outputs and controls.
- Essay questions: organize the answer with a direct introduction, developed paragraphs and a concise conclusion.

# Answer Guidance 6

## HOW TO CHECK YOUR WORK

- Definition questions: use precise subject vocabulary and avoid circular definitions.
- Explanation questions: show cause-and-effect or the sequence of reasoning.
- Comparison questions: use the same criteria for both sides and state similarities and differences.
- Application questions: use details from the scenario, data, system or example.
- Calculation questions: show formulas, substitutions, units and a reasonableness check.
- Evaluation questions: present relevant evidence, limitations and a supported conclusion.
- Technical questions: distinguish components, processes, inputs, outputs and controls.
- Essay questions: organize the answer with a direct introduction, developed paragraphs and a concise conclusion.

# MATHEMATICS INFORMATIQUE Reference

## QUICK REFERENCE

- Define key terms before applying them.
- Use symbols, notation and units consistently where relevant.
- Distinguish facts, assumptions, examples and conclusions.
- Use diagrams, tables, formulas or structured steps when they improve clarity.
- Check calculations and technical procedures independently where possible.
- For written analysis, connect each paragraph to the question.
- For practical work, document inputs, process, output, tests and corrections.
- Review weak lessons using spaced repetition and mixed practice.

# Final Revision Checklist

## BEFORE THE FINAL EXAMINATION

- ☐ I can define the core concepts accurately.
- ☐ I can explain methods and processes step by step.
- ☐ I can apply the subject to unfamiliar examples.
- ☐ I can identify common errors and correct them.
- ☐ I can compare related ideas using clear criteria.
- ☐ I can complete mixed practice without notes.
- ☐ I can check calculations, evidence, logic or technical results.
- ☐ I can write a concise, direct final conclusion.